SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING) Department of ArtificialIntelligence and Machine Learning

ASSESSMENT1 – Analytical Problem solving (5 QUESTIONS)

Course Code:	ITA05	**Course Title:**	**Computer Vision**
CO Assessed:	CO2	**Assessment:**	Assessment 1: Analytical Problem solving
Weightage:	35% of CIA	**Total Marks:**	(5 Questions × 10 Marks)
CO2 Statement:	***Apply image enhancement, filtering, segmentation techniques to analyze i edges and boundaries, and solve rea Computer Vision problems.***		
Student Name:	Vishnu Priya	**Reg. No.:**	192425309
Section / Batch:	B-Tech AIML	**Date:**	21/10/2026

Code of Conduct

I (NAME/ REG NO) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the Student:

QUESTIONS: 5 Total Marks: 50

Annexure A

Analytical Problems

Q1. A grayscale image contains the following pixel intensity values: 10, 10, 20, 20, 20, 30, 30, and 40. Construct the histogram by calculating the frequency of each intensity value. Calculate the cumulative distribution function (CDF) for the given image. Apply histogram equalization using the formula s = round((L − 1) × CDF), where L = 256. Determine the new intensity values and compare the contrast distribution before and after equalization.

Q2. A 3 × 3 image has the intensity values [[10, 20, 30], [20, 100, 20], [10, 20, 30]]. Apply a 3 × 3 averaging filter to calculate the new intensity value of the centre pixel. Replace the centre pixel with the calculated average while keeping the other pixels unchanged. Apply a median filter to the same neighbourhood and calculate the centre pixel value. Compare the results of the averaging and median filters and determine which filter better preserves image details in the presence of noise.

Q3. Consider a grayscale image with the intensity matrix [[50, 50, 50], [50, 100, 50], [50, 50, 50]]. Apply the Laplacian filter mask [[0, −1, 0], [−1, 4, −1], [0, −1, 0]] to calculate the response at the centre pixel. Use the calculated response to sharpen the centre pixel by adding it to the original intensity value. Determine the resulting centre pixel intensity after sharpening. Explain how the Laplacian filter enhances intensity transitions and fine details in the image.

Q4. A grayscale image contains the pixel intensity values [[40, 60, 120], [80, 150, 200], [30, 100, 180]]. Apply global thresholding with a threshold value of 100, assigning 1 to pixels greater than or equal to the threshold and 0 to the remaining pixels. Construct the resulting binary image. Calculate the number of foreground and background pixels after thresholding. Explain how the selected threshold separates objects from the background and identify one limitation of this method.

Q5. A grayscale image has the intensity matrix [[10, 10, 10], [10, 50, 90], [10, 50, 90]]. Apply the Sobel operator using the masks Gx = [[−1, 0, 1], [−2, 0, 2], [−1, 0, 1]] and Gy = [[−1, −2, −1], [0, 0, 0], [1, 2, 1]] to calculate the gradient components at the centre pixel. Calculate the gradient magnitude using $G = \sqrt{(Gx^2 + Gy^2)}$. Determine the dominant edge direction based on the calculated gradient components. Explain how the detected edge information can be used for boundary detection and image segmentation.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Content Accuracy	30	Highly accurate, indepth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach

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Use of Tools	20	Advanced tools, well analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	5	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	0	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Faculty In-charge	**Course Coordinator Program Director**
Signature:	Signature: Signature:
Date:	Date: Date:

Q1. Histogram, CDF and Histogram Equalization
Given pixel intensity values:

$$10,10,20,20,20,30,30,40$$

Total number of pixels, $N = 8$, and number of intensity levels, $L = 256$.
Step 1: Construct the histogram
The frequency of each intensity value is counted.

Intensity (r_k)	**Frequency (n_k)**	**Probability (n_k/N)**
10	2	0.25
20	3	0.375
30	2	0.25
40	1	0.125
Total	8	1.00

Step 2: Calculate the cumulative distribution function (CDF)
The CDF is the cumulative sum of the probabilities.

$$CDF(r_k) = \sum_{j=0}^{k} p\left(r_j\right)$$

Intensity	**Probability**	**CDF**
10	0.25	0.25
20	0.375	0.625
30	0.25	0.875
40	0.125	1.00

Step 3: Apply histogram equalization
The formula is

$$s = \text{round}[(L-1) \times CDF]$$

Since $L = 256$,

$$s = \text{round}(255 \times CDF)$$

Calculations:
- For intensity 10:

$$s = \text{round}(255 \times 0.25) = 64$$

- For intensity 20:

$$s = \text{round}(255 \times 0.625) = 159$$

- For intensity 30:

$$s = \text{round}(255 \times 0.875) = 223$$

- For intensity 40:

$$s = \text{round}(255 \times 1) = 255$$

Step 4: New intensity values

Original intensity	New intensity
10	64
10	64
20	159
20	159
20	159
30	223
30	223
40	255

The equalized image values are:

$$\boxed{64,64,159,159,159,223,223,255}$$

Comparison of contrast

- **Before equalization:** Pixel values lie between 10 and 40, giving a narrow intensity range.
- **After equalization:** Pixel values range from 64 to 255, spreading the intensities across a wider range.
- **Result:** The contrast between different intensity levels is enhanced, potentially making image features easier to distinguish.

Conclusion: Histogram equalization improves contrast by redistributing intensity values across a wider range.

Q2. Averaging Filter and Median Filter

Given image:

$$I = \begin{bmatrix} 10 & 20 & 30 \\ 20 & 100 & 20 \\ 10 & 20 & 30 \end{bmatrix}$$

We need to calculate the new centre pixel value using both filters.

Step 1: Apply the 3 × 3 averaging filter

The averaging filter calculates the mean of all nine pixels.

$$I_{\text{avg}} = \frac{\text{Sum of all pixels}}{9}$$

Sum of the pixels:

$$10 + 20 + 30 + 20 + 100 + 20 + 10 + 20 + 30 = 260$$

Therefore,

$$I_{\text{avg}} = \frac{260}{9} = 28.89$$

Rounding to the nearest integer:

$$\boxed{I_{\text{avg}} = 29}$$

The image after averaging is:

$$\boxed{\begin{bmatrix} 10 & 20 & 30 \\ 20 & 29 & 20 \\ 10 & 20 & 30 \end{bmatrix}}$$

Step 2: Apply the median filter
Arrange all nine values in ascending order:

$$10,10,20,20,20,20,30,30,100$$

Since there are nine values, the median is the fifth value.

$$\boxed{I_{\text{median}} = 20}$$

The image after median filtering is:

$$\boxed{\begin{bmatrix} 10 & 20 & 30 \\ 20 & 20 & 20 \\ 10 & 20 & 30 \end{bmatrix}}$$

Step 3: Compare the filters

Feature	**Averaging filter**	**Median filter**
Operation	Calculates the mean	Selects the middle sorted value
Centre pixel	29	20
Effect on noise	Reduces noise by smoothing	Removes isolated extreme values effectively
Detail preservation	Can blur edges and details	Generally preserves edges better

Conclusion: The median filter is better at preserving image details when noise consists of isolated extreme values, such as salt-and-pepper noise. The averaging filter smooths the image but is more affected by the unusually high value of 100.

Q3. Laplacian Filtering for Image Sharpening
Given image:

$$I = \begin{bmatrix} 50 & 50 & 50 \\ 50 & 100 & 50 \\ 50 & 50 & 50 \end{bmatrix}$$

Laplacian mask:

$$K = \begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

Step 1: Calculate the Laplacian response
Multiply the corresponding image and mask elements and add the results.

$$R = (0)(50) + (-1)(50) + (0)(50)$$

$$+ (-1)(50) + (4)(100) + (-1)(50)$$

$$+ (0)(50) + (-1)(50) + (0)(50)$$

Simplifying:

$$R = -50 - 50 + 400 - 50 - 50$$

$$\boxed{R = 200}$$

Step 2: Sharpen the centre pixel
According to the question, add the Laplacian response to the original centre intensity.

$$I_{\text{sharp}} = I_{\text{centre}} + R$$

$$I_{\text{sharp}} = 100 + 200$$

$$\boxed{I_{\text{sharp}} = 300}$$

The mathematical sharpened centre value is 300. However, in an 8-bit grayscale image, valid intensities range from 0 to 255, so the stored value would be clipped to 255.

Step 3: Explanation

- The Laplacian measures intensity changes around a pixel using its neighbouring pixels.
- Large responses indicate rapid intensity transitions, such as edges and fine details.
- Adding the response to the original image, as specified in this question, enhances the centre pixel's contrast relative to its surroundings.
- This makes boundaries and small details more prominent.

Conclusion: The Laplacian response is 200, and the calculated sharpened centre intensity is 300 (clipped to 255 for an 8-bit image).

Q4. Global Thresholding and Binary Image
Given image:

$$I = \begin{bmatrix} 40 & 60 & 120 \\ 80 & 150 & 200 \\ 30 & 100 & 180 \end{bmatrix}$$

Threshold value:

$$T = 100$$

The thresholding rule is

$$g(x,y) = \begin{cases} 1, & I(x,y) \geq 100 \\ 0, & I(x,y) < 100 \end{cases}$$

Step 1: Apply the threshold
Evaluate each pixel:

- $40 < 100 \Rightarrow 0$
- $60 < 100 \Rightarrow 0$
- $120 \geq 100 \Rightarrow 1$

- $80 < 100 \Rightarrow 0$
- $150 \geq 100 \Rightarrow 1$
- $200 \geq 100 \Rightarrow 1$
- $30 < 100 \Rightarrow 0$
- $100 \geq 100 \Rightarrow 1$
- $180 \geq 100 \Rightarrow 1$

Step 2: Construct the binary image

$$\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

Step 3: Count foreground and background pixels

Foreground pixels are assigned the value 1.

$$N_{\text{foreground}} = 5$$

Background pixels are assigned the value 0.

$$N_{\text{background}} = 4$$

Verification:

$$5 + 4 = 9 \text{ pixels}$$

Step 4: Explanation and limitation

- **Foreground:** Pixels with intensities greater than or equal to 100.
- **Background:** Pixels with intensities below 100.
- **Purpose:** The threshold separates brighter regions from darker regions, simplifying object detection and segmentation.

Limitation: Global thresholding uses one fixed threshold for the entire image. If illumination varies across the image or the foreground and background intensities overlap, it may misclassify pixels.

Conclusion: The binary image contains 5 foreground pixels and 4 background pixels.

Q5. Sobel Operator for Edge Detection

Given image:

$$I = \begin{bmatrix} 10 & 10 & 10 \\ 10 & 50 & 90 \\ 10 & 50 & 90 \end{bmatrix}$$

Sobel masks:

$$G_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$$

$$G_y = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

Step 1: Calculate the horizontal gradient G_x

Multiply corresponding elements of the image and G_x, then add.

$$G_x = (-1)(10) + (0)(10) + (1)(10)$$

$$+ (-2)(10) + (0)(50) + (2)(90)$$

$$+ (-1)(10) + (0)(50) + (1)(90)$$

Simplifying:

$$G_x = 0 + 160 + 80$$

$$\boxed{G_x = 240}$$

Step 2: Calculate the vertical gradient G_y

$$G_y = (-1)(10) + (-2)(10) + (-1)(10)$$

$$+ (0)(10) + (0)(50) + (0)(90)$$

$$+ (1)(10) + (2)(50) + (1)(90)$$

Simplifying:

$$G_y = -40 + 200$$

$$\boxed{G_y = 160}$$

Step 3: Calculate gradient magnitude

The gradient magnitude is

$$G = \sqrt{G_x^2 + G_y^2}$$

Substitute the calculated values:

$$G = \sqrt{240^2 + 160^2}$$

$$G = \sqrt{57600 + 25600}$$

$$G = \sqrt{83200}$$

$$\boxed{G \approx 288.44}$$

This indicates a strong intensity change around the centre pixel.

Step 4: Determine the dominant edge direction

The gradient angle is

$$\theta = \tan^{-1}\left(\frac{G_y}{G_x}\right)$$

$$\theta = \tan^{-1}\left(\frac{160}{240}\right)$$

$$\boxed{\theta \approx 33.69^\circ}$$

The gradient points approximately 33.69° from the positive horizontal axis. Since $G_x = 240$ is greater than $G_y = 160$, the horizontal gradient component is dominant.

Important distinction: The gradient direction indicates the direction of the greatest intensity increase. The edge itself is approximately perpendicular to the gradient, so the corresponding edge orientation is

approximately

$$33.69° + 90° = \boxed{123.69°}$$

Step 5: Applications of Sobel edge detection

- **Boundary detection:** Large gradient magnitudes indicate likely object boundaries.
- **Image segmentation:** Edge information can help separate an image into distinct regions or objects.
- **Feature extraction:** Detected edges can identify shapes, contours, and structural details.
- **Noise handling:** The Sobel operator includes weighted neighbourhood calculations that provide some smoothing, although it does not eliminate noise completely.

Conclusion: The gradient components are $G_x = 240$and $G_y = 160$, the gradient magnitude is approximately 288.44, and the gradient direction is approximately 33.69°. These values indicate a strong intensity transition useful for detecting edges and segmenting objects.





The following assessment has been discussed by the faculty and done at the class on 21st October 2026